

Boosting RF Fingerprint Identification with a Pre-trained Time-Series Encoder

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Abstract—Radio Frequency Fingerprint Identification (RFFI) serves as a critical physical-layer authentication technique for securing low-altitude wireless networks, particularly for Unmanned Aerial Vehicles (UAVs). By leveraging intrinsic hardware imperfections, RFFI provides a robust defense against identity spoofing. However, dynamic low-altitude environments present significant challenges, including weak signal fingerprints obscured by complex channel fading and the necessity to distinguish between known and unknown devices. To address these issues, this paper proposes a dual-stream hybrid framework based on the pre-trained TimeXer architecture. We treat RF signals as time-series data and transfer universal temporal features to the RFFI domain. The proposed framework integrates a Local-Global Temporal Fusion Network (LGT-Net) for closed-set recognition and a dual-stream inductive bias complementary model for unsupervised open-set anomaly detection. Experiments on real-world LTE datasets demonstrate that this approach achieves superior accuracy and generalization compared to non-pre-trained baselines, providing a viable security solution for dynamic low-altitude networks.

Index Terms—Low-altitude Economy, RF Fingerprint Identification, Pre-trained TimeXer, Anomaly Detection.

I. INTRODUCTION

The rapid expansion of the low-altitude economy, encompassing diverse applications such as Unmanned Aerial Vehicles (UAVs), aerial logistics, and Urban Air Mobility (UAM), has introduced unprecedented opportunities and security challenges to modern wireless communication systems. In these dynamic environments, ensuring reliable authentication is critical to maintaining the integrity of control and data links. Radio Frequency Fingerprint Identification (RFFI) has emerged as a promising physical-layer authentication technique. By leveraging intrinsic hardware imperfections—such as amplifier non-linearity and phase noise—RFFI creates unique electromagnetic signatures for device identification [1]. This non-intrusive and difficult-to-forge technology serves as a key enabler for secure smart low-altitude infrastructures [2], [3]. However, the practical deployment of RFFI in low-altitude networks is impeded by the complex nature of the wireless environment and the limitations of existing identification methodologies. Traditional RFFI approaches typically rely on manually extracted statistical features coupled with classical classifiers such as support vector machines (SVM) [4]. These methods depend heavily on expert domain knowledge and

often fail to capture the deep, non-linear abstractions necessary to distinguish devices in complex channel conditions. To address these deficiencies, deep learning techniques have been widely adopted [5]. ResNet-based architectures have been applied to massive MIMO systems [6], while attention mechanisms and Transformers have been utilized to model temporal dependencies in signal sequences [7]. Furthermore, generative approaches, such as diffusion models, have been introduced to augment training data and improve robustness [8].

Despite these advances, current deep learning solutions exhibit significant limitations when applied to large-scale and open environments. Most existing neural networks, including standard CNNs and Transformers, predominantly focus on extracting local waveform features or short-term dependencies. They often lack the capacity to model the global temporal evolution and long-range dependencies of the signal. This deficiency leads to two critical unresolved challenges in real-world low-altitude networks:

- **Scalability in Multi-Device Scenarios:** As the number of accessing terminals increases, the decision boundaries between devices become increasingly complex. Models that rely solely on local features struggle to maintain high recognition accuracy in dense, multi-user environments due to insufficient feature discriminability.
- **Unknown Device Detection:** In open-set scenarios, the system must identify authorized users while rejecting unauthorized or unknown devices (out-of-distribution, OOD). Existing models often overfit the training distribution and fail to generalize, leading to severe performance degradation when exposed to anomalous signals [9].

To overcome these bottlenecks, this paper proposes a novel RFFI framework that incorporates TimeXer, a state-of-the-art time-series encoder pre-trained on diverse datasets [10]. We hypothesize that RF signals, when treated as one-dimensional time series, share universal temporal properties with general time series data. Using temporal encoders such as TimesFM [11] and TimeXer [10], [12], we can transfer universal feature extraction capabilities to the RFFI domain. This allows the model to capture global signal evolution patterns that are robust against noise and channel variations, addressing the limitations of local modeling found in traditional frameworks [13]–[15]. Our main contributions are summarized as follows:

- We propose the Local-Global Temporal Fusion Network

(LGT-Net) for closed-set recognition. This architecture fuses local waveform features extracted by 1D-CNNs with global temporal representations from a pre-trained TimeXer. This combination significantly improves identification accuracy in multi-user scenarios by integrating fine-grained local details with macroscopic temporal stability.

- We develop a Dual-Stream Inductive Bias Complementary (DSIBC) model for open-set recognition. This unsupervised approach integrates local contrastive learning (via TS2Vec) with global distribution modeling (via TimeXer) to effectively detect and reject unknown devices based on deviations in both local consistency and global temporal patterns.
- We validate the framework on a real-world dataset of LTE uplink signals collected from 100 distinct terminals using an active interrogation mechanism. Experimental results confirm that our method outperforms standard baselines in both classification accuracy and anomaly detection capabilities.

II. CLOSED-SET RECOGNITION METHOD

A. Data Modeling and Preprocessing

To address the challenge of device identification in dynamic low-altitude environments, this paper employs an active interrogation mechanism for signal acquisition. In this scheme, the base station transmits inducing signals to trigger the target terminal's response, while the receiver synchronously captures the uplink feedback. This mechanism allows for the construction of a labeled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where $\mathbf{x}_i$ denotes the i -th RF signal sample and $y_i \in \{1, \dots, K\}$ represents the class label corresponding to the terminal's International Mobile Subscriber Identity (IMSI).

The raw signals captured at the receiver are complex-valued IQ sequences. To mitigate the effects of channel fading, noise, and receiver gain variations, the continuous signals are sliced into non-overlapping segments of fixed length $L = 4096$. Subsequently, Z-Score normalization is applied to each sample to standardize the data distribution [16]. Unlike traditional methods that utilize only amplitude information, we map the complex signals into a dual-channel real-valued tensor to preserve both energy variations and phase fingerprints. Consequently, each processed sample is represented as $\mathbf{x}_i \in \mathbb{R}^{2 \times L}$, where the two channels correspond to the In-phase (I) and Quadrature (Q) components, respectively. This preprocessing strategy retains local waveform details while standardizing input scales for the neural network.

B. LGT-Net Dual-Stream Hybrid Network Model

To effectively model the multi-scale temporal characteristics of RF fingerprints, we propose the Local-Global Temporal Fusion Network (LGT-Net), as shown in Fig. 1. This architecture integrates a pre-trained time-series encoder with a one-dimensional convolutional network (1D-CNN). The model achieves collaborative representation of discriminative features by modeling local high-resolution waveform characteristics

and global long-range temporal dependencies in parallel. The overall architecture consists of two streams:

1) *Local Stream*: For the input signal $\mathbf{x}_i$, the local branch extracts short-term waveform structural features through a large-kernel 1D-CNN. This is followed by global pooling to obtain the local representation $\mathbf{z}_{\text{cnn}} = f_{\text{cnn}}(\mathbf{x}_i)$. This branch is designed to be highly sensitive to fine-grained transient distortions caused by hardware non-idealities.

2) *Global Stream*: The global branch first compresses the sequence length via adaptive average pooling to reduce computational complexity. It then employs the pre-trained TimeXer encoder to process the compressed sequence. Through patch-level self-attention mechanisms, the encoder models long-range dependencies across time segments, yielding a stable global temporal representation $\mathbf{z}_{\text{tx}} = f_{\text{TimeXer}}(\mathbf{x}_i')$. Benefiting from the temporal priors acquired during pre-training on large-scale time-series datasets, this representation maintains strong discriminative consistency under varying channel conditions.

3) *Feature Fusion and Classification*: The features from the two streams are concatenated along the feature dimension to form the fused representation: $\mathbf{z}_{\text{fused}} = [\mathbf{z}_{\text{cnn}} \parallel \mathbf{z}_{\text{tx}}]$. This fused vector is then fed into a classification head. During training, the Focal Loss is employed to address potential class imbalances and focus learning on hard examples:

$$\mathcal{L}_{\text{cls}} = -(1 - p_t)^\gamma \log(p_t) \quad (1)$$

where p_t is the predicted probability of the true class, and γ is the focusing parameter. By fusing the local inductive bias of convolutional networks with the global temporal modeling capability of the pre-trained TimeXer, the LGT-Net achieves robust performance in closed-set RFFI tasks.

III. OPEN-SET RECOGNITION METHOD

A. Data Modeling and Tensor Construction

In real-world mobile communication environments, the receiver inevitably encounters signals from unknown terminals or anomalous devices due to the isotropic propagation characteristics of radio frequency signals. This necessitates the ability to detect OOD samples. Following the active interrogation mechanism described in Section II, the data is partitioned at the terminal level to strictly adhere to the unsupervised anomaly detection setting.

Formally, the known device dataset is defined as $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where $y_i \in \{1, \dots, K\}$. The unknown device (OOD) dataset is denoted as $\bar{\mathcal{D}} = \{(\bar{\mathbf{x}}_j, \bar{y}_j)\}_{j=1}^{\bar{N}}$, with $\bar{y}_j \notin \{1, \dots, K\}$. The training set $\mathcal{D}_{\text{train}} \subset \mathcal{D}$ contains only normal samples from known categories, while the evaluation is performed on $\mathcal{D}_{\text{test}} \cup \bar{\mathcal{D}}$. The raw signals are complex variable-length IQ sequences. To achieve uniform modeling, we employ a time window of length $L = 4096$. Replication padding is applied to sequences of insufficient length to avoid spectral artifacts introduced by zero-padding. Unlike the closed-set scenario, OOD detection benefits from a richer feature set. Therefore, the signals are mapped into a multi-channel real-valued tensor $\mathbf{X} \in \mathbb{R}^{C \times L}$, where C represents the channels including In-phase (I), Quadrature (Q), Amplitude,

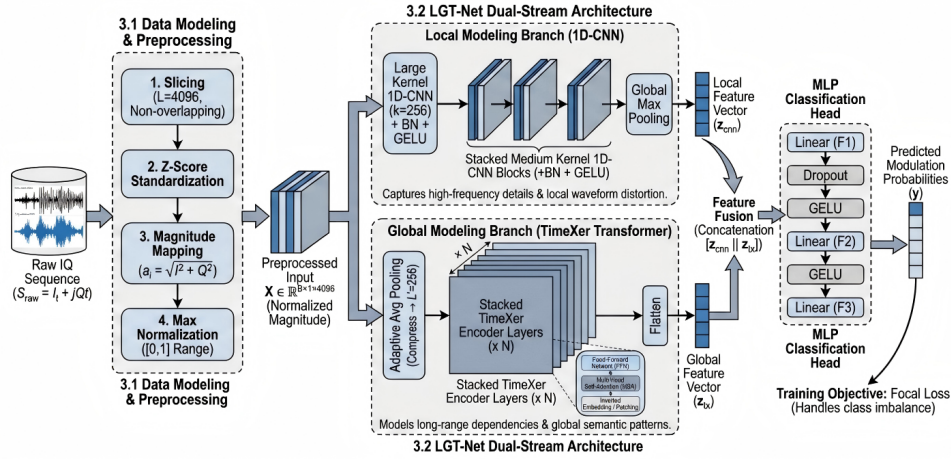


Fig. 1. The LGT-Net Dual-Stream Hybrid Network Model.

and Phase information. Considering that OOD detection is highly sensitive to absolute energy variations, we perform global standardization based on statistics from the normal training samples: $\mathbf{X}' = \frac{\mathbf{X} - \mu_{\mathcal{T}}}{\sigma_{\mathcal{T}}}$, where $\mu_{\mathcal{T}}$ and $\sigma_{\mathcal{T}}$ are the channel-wise mean and standard deviation calculated over the training set $\mathcal{D}_{train}$. This approach unifies feature scales while explicitly retaining energy shifts caused by hardware differences and link state variations, providing effective criteria for subsequent anomaly determination.

B. Dual-Stream Complementary Anomaly Detection Model

To fully exploit anomaly patterns in RF signals at different time scales, we propose a dual-stream unsupervised anomaly detection model, as shown in Fig. 2. This model integrates the pre-trained TimeXer architecture with the TS2Vec temporal contrastive learning framework [17]. The overall structure leverages complementary inductive biases: local consistency verification and global distribution matching.

1) *Local Branch (TS2Vec)*: The local branch is based on the TS2Vec architecture, which extracts multi-scale local features through residual dilated convolutions. This branch is optimized using hierarchical contrastive learning to capture contextual consistency. The core objective is formulated as:

$$\mathcal{L}_{\text{local}} = -\log \frac{\exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+)/\tau)}{\sum_j \exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_j)/\tau)} \quad (2)$$

where $\mathbf{h}_i$ and $\mathbf{h}_i^+$ represent positive sample pairs constructed through random cropping and temporal masking, $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity function, and τ is the temperature parameter. This objective encourages the model to maintain high sensitivity to local waveform structural variations, ensuring that anomalies disrupting local signal continuity are detected.

2) *Global Branch (TimeXer)*: The global branch incorporates the pre-trained TimeXer to encode the complete RF time series, yielding the global representation $\mathbf{g}_i = f_{\text{TimeXer}}(\mathbf{x}_i)$. By characterizing the statistical distribution of known devices

within this feature space, the model explicitly learns long-term temporal structures and energy evolution patterns. This enhances the ability to identify OOD samples that may locally resemble valid signals but deviate in global temporal behavior.

3) *Anomaly Scoring and Fusion*: During the inference phase, the model generates two anomaly scores: S_{local} , reflecting the degree of local consistency disruption, and S_{global} , representing the deviation from the global feature distribution of known devices. These scores are fused via linear weighting to obtain the final anomaly score: $S = \lambda S_{\text{local}} + (1 - \lambda) S_{\text{global}}$, where λ is a balancing hyperparameter. Furthermore, statistical aggregation is performed over multiple samples from the same device to achieve stable device-level open-set determination. This approach integrates the complementary advantages of local contrastive learning and pre-trained global temporal modeling, enabling effective detection of unknown RF devices without requiring exposure to OOD samples during training.

IV. EXPERIMENTS

To validate the proposed method for cellular-connected UAV applications, we utilize a dataset of real-world 4G LTE uplink signals. The rationale for using this dataset is twofold. First, cellular-connected UAVs function as aerial User Equipment (UE), utilizing identical RF transceiver chipsets and waveform standards (SC-FDMA/OFDMA) as ground terminals. Consequently, the hardware impairments (e.g., I/Q imbalance, phase noise) manifested in this dataset are representative of those found in aerial nodes. Second, LTE uplink signals provide a standardized, challenging environment with realistic channel fading and interference, serving as a valid proxy for evaluating physical layer identification in low-altitude networks [18].

Data was collected from 100 distinct LTE terminals across two frequency bands: B1 (2100 MHz) and B3 (1800 MHz). We extract the Demodulation Reference Signal (DMRS) from the uplink subframes as the fingerprint feature, as it provides a consistent pilot sequence ideal for identification [19].

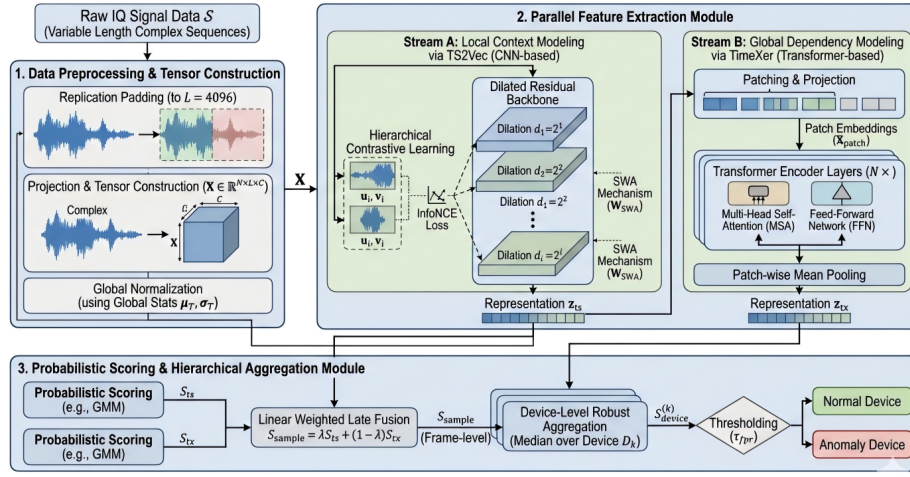


Fig. 2. The Unsupervised Anomaly Detection Model.

A. Closed-Set Recognition Performance

We compare LGT-Net against a standard 1D-CNN and a DNN baseline. The dataset is split 4:1 for training and testing. As shown in Table I, LGT-Net achieves the highest accuracy on both bands (96% on B3, 97% on B1). The performance gain over 1D-CNN confirms that the pre-trained TimeXer encoder successfully captures subtle temporal dependencies that purely convolutional architectures miss. This suggests that transferring knowledge from general time-series domains effectively enhances RF fingerprinting, particularly in distinguishing devices with highly similar hardware profiles.

TABLE I
CLOSED-SET RECOGNITION ACCURACY (100 DEVICES)

Dataset	LGT-Net	1D-CNN	DNN
B3 (1800 MHz)	0.96	0.95	0.91
B1 (2100 MHz)	0.97	0.95	0.94

B. Open-Set Recognition Performance

For open-set evaluation, we designate a subset of devices as “Known” and the rest as “Unknown”. We compare our Dual-Stream Inductive Bias Complementary (DSIBC) model against Generative Adversarial Networks (GAN), TS2Vec, and a TimeXer+GAN hybrid. The metrics are the open-set detection accuracy and the area under the receiver operating characteristic curve (AUROC).

Table II presents the results under different known-device scales (10 and 70 devices) and frequency conditions. DSIBC consistently outperforms the baselines. Notably, at 1250 MHz with 70 devices, it maintains an accuracy of 0.87, whereas the GAN-based approach drops to 0.57. The pure TS2Vec model performs well (0.92-0.93) but is surpassed by DSIBC (0.95), indicating that adding the global TimeXer branch provides critical complementary information. Fig. 3 illustrates that the DSIBC curve encloses the largest area, confirming superior sensitivity to unknown signals.

TABLE II
OPEN-SET DETECTION ACCURACY

Scale	Freq.	DSIBC (Ours)	GAN	TS2Vec	TimeXer+GAN
10 Devices	1900M	0.95	0.88	0.93	0.81
	1250M	0.95	0.60	0.92	0.82
70 Devices	1900M	0.88	0.56	0.79	0.69
	1250M	0.87	0.57	0.78	0.70

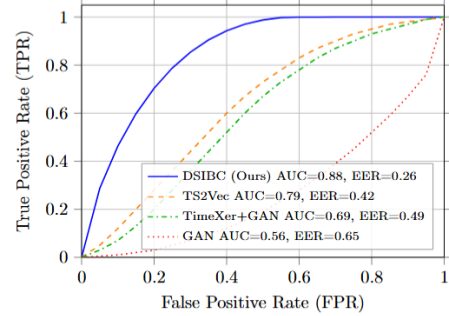


Fig. 3. ROC Curve Comparison of Anomaly Detection Methods.

V. CONCLUSION

This paper proposes a local-global dual-stream RF recognition framework based on the pre-trained TimeXer, and validates its effectiveness in both closed-set and open-set tasks. In closed-set scenarios, LGT-Net achieves high-precision, cross-band stable device identification; in open-set scenarios, the dual-stream unsupervised anomaly detection model significantly enhances the rejection capability for unknown devices. Experimental results demonstrate that the pre-trained time-series model can substantially improve the stability and generalization ability of RF feature representations, enabling the proposed method to provide reliable recognition and detection performance in complex wireless environments.

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